

Design and Characterization of Silicon Photonic Mach–Zehnder Interferometers with Compact Modeling and Corner Analysis

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Abstract—This report presents the design, fabrication, and experimental characterization of Mach–Zehnder interferometers (MZIs) on a silicon-on-insulator (SOI) platform using strip waveguides. A complete design workflow is established, spanning waveguide-level eigenmode simulation, compact model extraction, and circuit-level simulation. A linearized analysis based on the reciprocal path length difference is employed to relate the free spectral range (FSR) to the waveguide group index. A corner analysis is performed to quantify the sensitivity of the group index and FSR to ± 10 nm manufacturing variations in waveguide width and thickness. Finally, eight fabricated MZI devices are characterized by automated optical measurement, and the extracted FSR and group index are compared against simulation. The measured group index of $n_g = 4.32$ agrees with the simulated value of 4.20 to within 2.7%, and two nominally identical devices exhibit FSR repeatability within 0.7%, confirming the consistency of the design flow and the uniformity of the fabrication process.

Index Terms—Silicon Photonics, Silicon-on-Insulator (SOI), Mach-Zehnder Interferometer (MZI), Free Spectral Range, Corner Analysis, Passive Components.

I. INTRODUCTION

The Mach-Zehnder Interferometer (MZI) is a fundamental component in silicon photonics, finding extensive applications in optical modulators, optical switches, and sensors. This work establishes a comprehensive MZI design workflow spanning from waveguide modeling to circuit-level simulation, manufacturing-variability analysis, and experimental verification. A linearized Free Spectral Range (FSR) analysis method is introduced to provide a consistency check for the extracted group index, and the simulated behavior is validated against measurements from fabricated devices.

II. THEORY

A. MZI Transfer Function

Consider an unbalanced Mach-Zehnder Interferometer (MZI) where the input light is split into two arms by a Y-branch. After propagating through waveguides of lengths L_1 and L_2 , the light is recombined. The transfer function $T_{MZI}(\lambda)$ can be expressed as [1]:

$$T_{MZI}(\lambda) = \frac{1}{2}[1 + \cos(\beta\Delta L)] \quad (1)$$

where $\beta = 2\pi n_{eff}/\lambda$ is the propagation constant and $\Delta L = L_2 - L_1$ represents the path length difference. This expression

assumes ideal 50:50 power splitting and negligible imbalance or excess loss, which is sufficient for FSR analysis.

B. Free Spectral Range (FSR)

The Free Spectral Range (FSR) is defined as the wavelength spacing between two adjacent transmission peaks:

$$FSR = \frac{\lambda^2}{n_g \Delta L} \quad (2)$$

where n_g is the group index, defined as $n_g = n_{eff} - \lambda \frac{dn_{eff}}{d\lambda}$.

C. Linear Relationship between FSR and $1/\Delta L$

In this work, λ is set to 1550 nm, where the compact model is defined and the dispersion within the considered bandwidth is sufficiently small. By rewriting the FSR equation, we obtain a linear relationship:

$$FSR = \left(\frac{\lambda^2}{n_g} \right) \cdot \left(\frac{1}{\Delta L} \right) \quad (3)$$

This representation linearizes the relationship between FSR and the reciprocal of the path length difference ($1/\Delta L$), where the slope is λ^2/n_g . This approach is advantageous for:

- **Data Visualization:** Linear plots are more intuitive than hyperbolic ones.
- **Parameter Extraction:** The group index n_g can be estimated through linear fitting.
- **Model Accuracy:** The linear fitting provides a quantitative measure of internal consistency between the model and the simulation or measurement results.

III. WAVEGUIDE MODELING

A. Simulation Settings

A strip waveguide with cross-sectional dimensions of $500 \text{ nm} \times 220 \text{ nm}$ was analyzed using the Lumerical MODE Finite Difference Eigenmode (FDE) solver. All waveguide parameters reported in this section are extracted at $\lambda = 1550 \text{ nm}$. The simulation parameters were configured as follows:

- **FDE Region:** $5.0 \mu\text{m}$ (X-span) $\times$ $1.7 \mu\text{m}$ (Y-span).
- **Mesh Grid:** 100×100 .
- **Boundary Conditions:** Metal boundaries.
- **Materials:** Si and SiO₂ using the SiEPIC EBeam PDK material models with multi-coefficient fitting over the wavelength range of (1.2 to 1.8 μm), the fitting tolerances are set to 0.1 and 0.001.

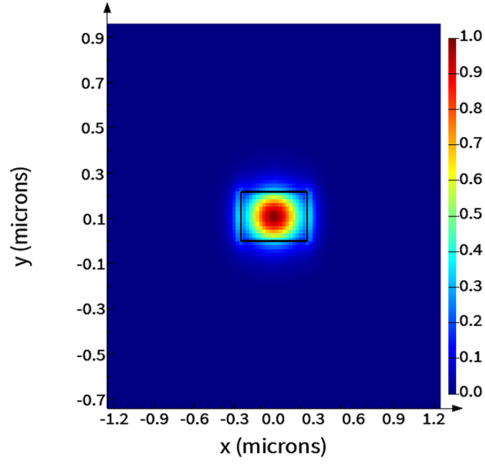


Fig. 1. Electric field intensity distribution of the fundamental TE mode at 1550 nm.

B. Modal Characteristics

The modal properties at the operating wavelength of 1550 nm are summarized in Table I. The fundamental TE mode field distribution is shown in Fig. 1.

TABLE I
SIMULATED WAVEGUIDE MODAL PARAMETERS AT A WAVELENGTH OF 1550 NM.

Parameter	Value
n_{eff}	2.447
n_g	4.203
Loss (dB/cm)	4.4×10^{-4}

C. Dispersion Characteristics

Wavelength sweeping from 1500 nm to 1600 nm revealed the following trends. The wavelength-dependent behaviors of n_{eff} and n_g are illustrated in Fig. 2 and Fig. 3, respectively.

- n_{eff} decreases with increasing wavelength, exhibiting normal dispersion.
- n_g increases slightly with wavelength (approximately 4.197 to 4.210).

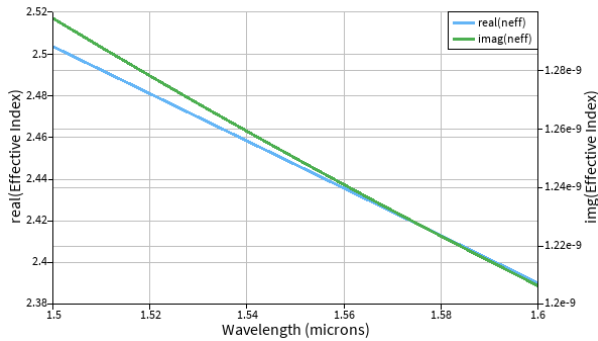


Fig. 2. Simulated effective index (n_{eff}) of the SOI strip waveguide as a function of wavelength.

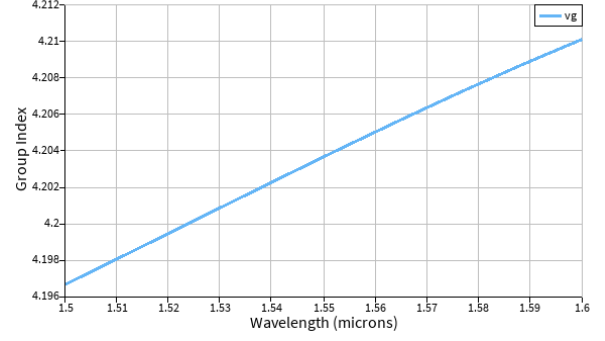


Fig. 3. Simulated group index (n_g) of the SOI waveguide as a function of wavelength.

D. Compact Model

To facilitate circuit-level simulation, a second-order Taylor expansion was used to establish a compact model for n_{eff} at a center wavelength of 1.55 μm . This model was implemented using a Lumerical MODE script which is

$$n_{eff}(\lambda) = 2.447 - 1.1334(\lambda - 1.55) - 0.04394(\lambda - 1.55)^2 \quad (4)$$

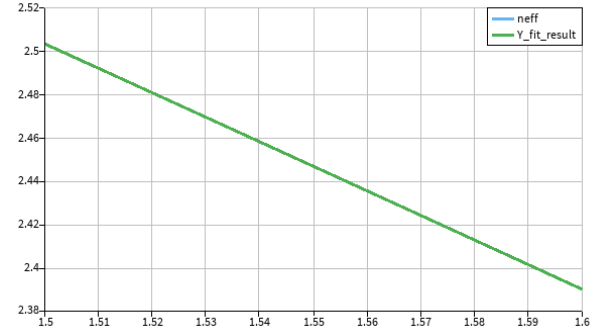


Fig. 4. Second-order Taylor expansion compact model of the SOI waveguide effective index extracted from MODE simulations.

IV. MZI DESIGN AND SIMULATION

A. Circuit Implementation

The MZI circuit was implemented in Lumerical INTERCONNECT [2] using the compact model derived above for the waveguides. For the input/output coupling and power splitting, standard passive components from the SiEPIC EBeam PDK [3] were utilized:

- **Grating Couplers (GCs):** Standard TE-mode focusing grating couplers were used to couple light between the optical fiber and the chip.
- **Y-Branched:** Optimized adiabatic Y-branch splitters were used to ensure 50:50 power splitting with minimal insertion loss.
- **Optical Network Analyzer (ONA):** Used for frequency domain characterization within the simulation environment.

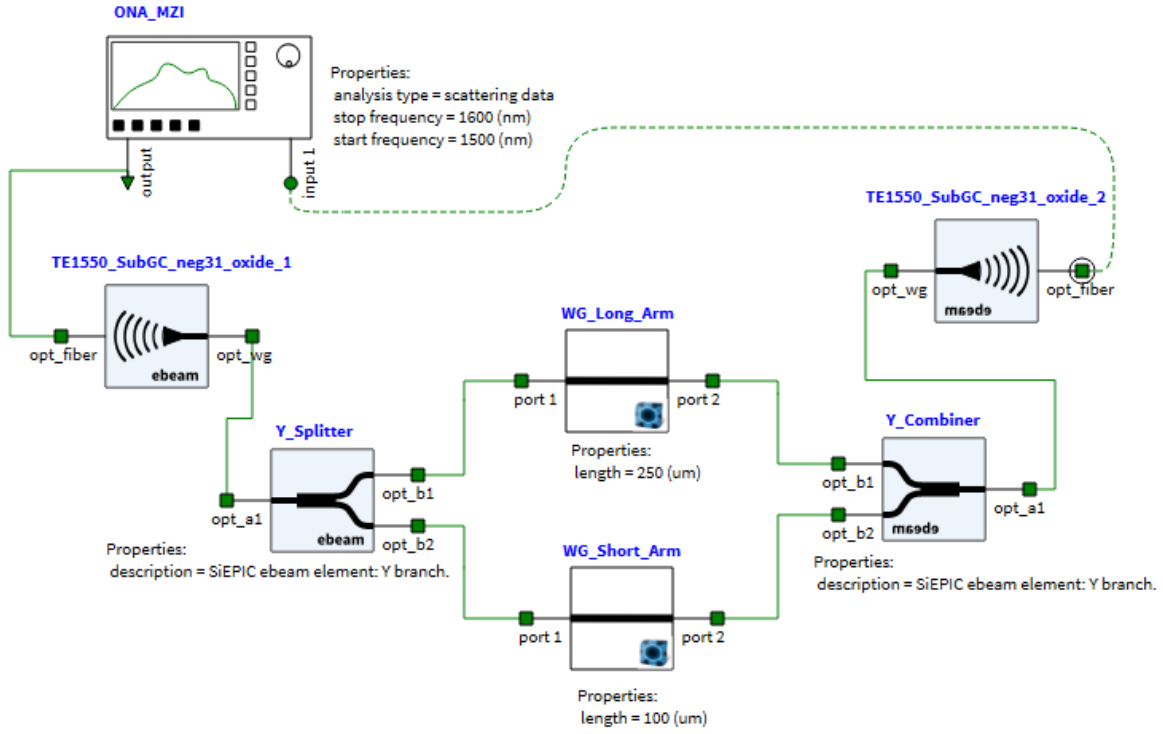


Fig. 5. MZI circuit schematic implemented in INTERCONNECT. The configuration shown corresponds to $\Delta L = 150 \mu\text{m}$ (long arm: $250 \mu\text{m}$, short arm: $100 \mu\text{m}$) as a representative example. Other MZI variants were simulated by adjusting the arm lengths accordingly.

B. Design Specifications and Results

Table II summarizes the design parameters for the MZI configurations with different path length differences (ΔL) and compares the theoretical FSR with simulation results. The fabricated chip integrates eight MZI devices: a balanced reference ($\Delta L = 0$), an FSR sweep ($\Delta L = 50, 100, 150, 163, 250 \mu\text{m}$), and a pair of nominally identical devices ($\Delta L = 150 \mu\text{m}$) for repeatability assessment.

TABLE II
REPRESENTATIVE MZI DESIGN PARAMETERS AND SIMULATION RESULTS. THEORETICAL VALUES WERE CALCULATED USING EQ. (2), WHILE SIMULATED VALUES WERE OBTAINED FROM INTERCONNECT.

$\Delta L (\mu\text{m})$	Theoretical FSR (nm)	Simulated FSR (nm)	Error (%)
50	11.43	11.32	-1.0
100	5.72	5.68	-0.7
150	3.81	3.79	-0.5
163	3.51	3.49	-0.6
250	2.29	2.28	-0.4

C. FSR Characteristics and Group Index Verification

To verify the consistency of the MZI model, the relationship between the FSR and the inverse path length difference ($1/\Delta L$) was linearized. A linear regression performed on the simulated data points using MATLAB yielded a coefficient of determination $R^2 = 0.99999$ and an extracted group index of $n_g = 4.21$.

To validate the extracted group index, the results obtained from two independent methods are compared:

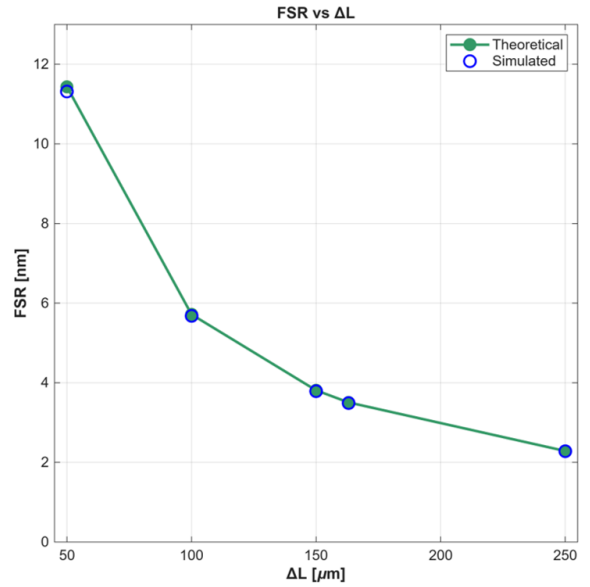


Fig. 6. Linearized free spectral range (FSR) as a function of $1/\Delta L$ obtained from the simulated MZI devices.

- **Method 1 (MODE Direct Calculation):** The group index was directly extracted from the FDE frequency sweep at $\lambda = 1550 \text{ nm}$ using $n_g = c/v_g$, where v_g is the group velocity computed by Lumerical MODE, yielding $n_g = 4.203$.

- **Method 2 (INTERCONNECT FSR Fitting):** Five MZI circuits with $\Delta L = 50\text{--}250\ \mu\text{m}$ were simulated in INTERCONNECT using the compact model. The FSR of each device was extracted from the simulated transmission spectrum, and n_g was obtained from the slope of the FSR vs. $1/\Delta L$ linear fit, yielding $n_g = 4.21$.
- **Relative Difference:** $< 0.5\%$

The small difference confirms that the compact model derived from MODE correctly represents the MZI behavior in INTERCONNECT.

D. Transmission Characteristics and Loss Analysis

The transmission spectra for the $\Delta L = 150\ \mu\text{m}$ configuration were analyzed (Fig. 7). Key observations include:

- **Ideal MZI (without Grating Couplers):** Peak transmission $\approx 0\ \text{dB}$, with an extinction ratio (ER) exceeding 60 dB.
- **MZI with Grating Couplers (GCs):** Peak transmission $\approx -6\ \text{dB}$, primarily due to the insertion loss of two GCs.
- **Loss Breakdown:** The total loss is dominated by the GCs ($\sim 6\ \text{dB}$ total), while the Y-branch splitters and waveguide propagation have a minimal impact on the overall transmission level.

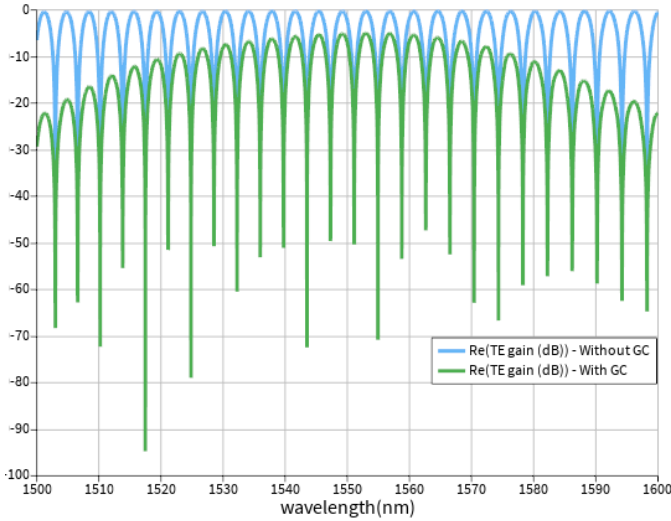


Fig. 7. Simulated transmission spectrum of the MZI with $\Delta L = 150\ \mu\text{m}$.

V. CORNER ANALYSIS

To assess the robustness of the design against manufacturing variability, a corner analysis was performed. A variation of $\pm 10\ \text{nm}$ in both width and thickness was chosen as a reasonable approximation of typical process fluctuations in electron-beam lithography, providing a conservative estimate of design robustness. The waveguide width and thickness were independently varied around the nominal $500 \times 220\ \text{nm}$ cross-section, producing four process corners in addition to the nominal case. For each corner, the group index was extracted at $\lambda = 1550\ \text{nm}$ using the MODE FDE solver. The results are summarized in Table III.

TABLE III
WAVEGUIDE MODAL PARAMETERS AT PROCESS CORNERS ($\lambda = 1550\ \text{nm}$).
 W : WIDTH; T : THICKNESS.

Case	$W\ (\text{nm})$	$T\ (\text{nm})$	n_{eff}	n_g
Nominal	500	220	2.447	4.203
C1 (+,+)	510	230	2.497	4.187
C2 (+,-)	510	210	2.419	4.167
C3 (-,+)	490	230	2.443	4.220
C4 (-,-)	490	210	2.385	4.199

The group index ranged from 4.167 (C2, wide and thin) to 4.220 (C3, narrow and thick), corresponding to a variation of $\pm 0.62\%$ relative to the nominal value. The resulting FSR variation across all MZI designs is therefore approximately $\pm 0.6\%$. Notably, the thickness variation has a stronger and opposite effect on n_g compared with the width variation, so the extrema occur at the “narrow + thick” (C3) and “wide + thin” (C2) corners. This indicates that maintaining tight thickness control is more critical than width control for FSR-sensitive applications.

VI. LAYOUT DESIGN

The physical layout was designed for fabrication and automated testing. The design features include:

- **Device Array:** Integration of eight MZI devices, comprising an FSR sweep ($\Delta L = 50$ to $250\ \mu\text{m}$), a balanced reference ($\Delta L = 0$), and a pair of identical $\Delta L = 150\ \mu\text{m}$ devices for repeatability assessment. The largest-imbalance device employs a meandered long arm to minimize footprint.
- **Grating Coupler Pitch:** Pitch of $127\ \mu\text{m}$ was implemented to ensure compatibility with fiber arrays.
- **Bending Radius:** The waveguide bends required a radius sufficiently large to maintain consistency with the straight-waveguide compact model used in our circuit simulations.
- **Testing Orientation:** All grating couplers are oriented to the right to facilitate automated testing and alignment.

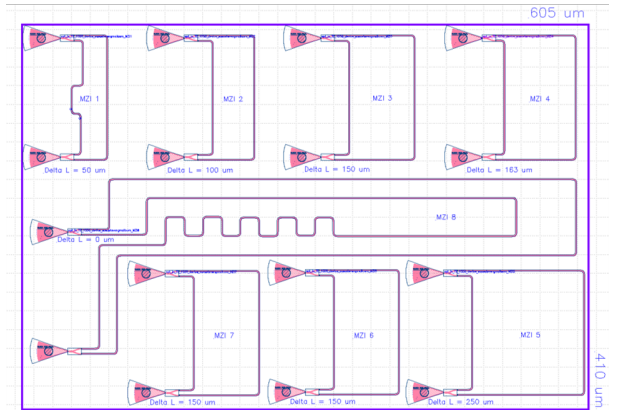


Fig. 8. Layout of the eight MZI devices, including the FSR sweep, the balanced reference, and the identical-device pair.

VII. FABRICATION AND MEASUREMENT RESULTS

A. Measurement Setup

The fabricated devices were characterized using an automated fiber-alignment measurement system. A tunable laser (HP81680A) swept from 1460 nm to 1580 nm in steps of 0.008 nm at an output power of -1.74 dBm. The transmitted power was recorded by a photodetector (HP81635A) through single-mode fiber pigtails coupled via the on-chip grating couplers.

B. Transmission Spectra

Fig. 9 shows the measured transmission spectra of all eight MZI devices. Clear interference fringes are observed across the C-band, modulated by the broadband grating-coupler envelope centered near 1550 nm. The balanced reference (MZI8, $\Delta L = 0$) exhibits no fringes, as expected.

C. FSR Extraction and Group Index Verification

Transmission peaks were identified using the MATLAB `findpeaks` function, and the FSR of each device was extracted via linear regression of consecutive peak wavelengths. Table IV compares the measured FSR with simulation. The measured values agree with simulation to within 1.7% for all devices except MZI1, which exhibits the largest FSR (~ 11 nm) and therefore only four detectable peaks within the measurement bandwidth, reducing its fit reliability.

TABLE IV
MEASURED FSR AND EXTRACTED GROUP INDEX COMPARED WITH
SIMULATION ($\lambda = 1550$ NM).

Device	ΔL (μm)	FSR _{sim} (nm)	FSR _{meas} (nm)	Error (%)	$n_{g,\text{exp}}$
MZI1	50	11.432	10.927	-4.4^*	4.397
MZI2	100	5.716	5.621	-1.7	4.274
MZI3	150	3.811	3.785	-0.7	4.232
MZI4	163	3.507	3.486	-0.6	4.229
MZI5	250	2.286	2.277	-0.4	4.221
MZI6	150	3.811	3.765	-1.2	4.254
MZI7	150	3.811	3.791	-0.5	4.225

* Only 4 peaks detected; excluded from the linear fit.

Fig. 10 shows the linearized FSR versus $1/\Delta L$ relationship for both simulated and measured data. A linear regression of the measured devices (excluding MZI1) yields an experimentally extracted group index of $n_g = 4.32$ ($R^2 = 0.9999$), compared with the simulated value of 4.203, a difference of 2.7%. Devices with larger ΔL (MZI3–MZI5, MZI7) show closer agreement with simulation ($< 1.2\%$ error), since the greater number of detectable peaks reduces the uncertainty in the FSR estimate. The measured group index exceeding the nominal value is consistent with a fabricated waveguide cross-section slightly narrower or thinner than designed, in line with the “narrow” corners of Table III.

D. Repeatability Analysis

MZI6 and MZI7 are nominally identical ($\Delta L = 150 \mu\text{m}$) and were included to assess fabrication uniformity and measurement repeatability. The measured FSR values differ by

only 0.026 nm (0.69%), and the corresponding group indices differ by 0.70%, confirming excellent process uniformity across the chip. The overlaid spectra are shown in Fig. 11.

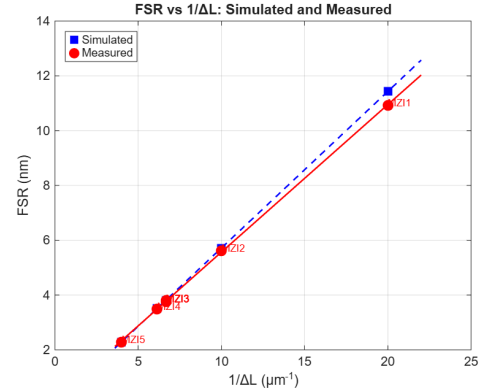


Fig. 10. Measured and simulated FSR as a function of $1/\Delta L$. The experimental group index is extracted from the slope of the linear fit (MZI1 excluded).

E. Insertion Loss Analysis

The balanced reference (MZI8, $\Delta L = 0$) provides a direct measure of the system insertion loss, since it contains no interferometric contribution. With a laser output of -1.74 dBm and a measured baseline of approximately -31.3 dBm near 1550 nm, the total fiber-to-fiber loss is approximately 29.5 dB, corresponding to roughly 14.8 dB per grating coupler. This value is higher than the simulated GC insertion loss, which is expected for automated measurement systems where fiber alignment is not manually optimized. Grating couplers are highly sensitive to fiber angle and position, and non-ideal alignment in automated setups typically results in additional coupling loss beyond the designed value. The spectral envelope shape of MZI8 remains consistent with the simulated GC response, confirming that the elevated loss reflects measurement conditions rather than a deficiency in the GC design.

VIII. RESULTS AND DISCUSSION

The proposed design flow was validated across simulation, manufacturing-variability analysis, and experiment:

- 1) **Compact Model:** The second-order Taylor expansion achieved an excellent fit for the effective index (n_{eff}) across the 1500 to 1600 nm range, and reproduced the MZI FSR in INTERCONNECT to within 1%.
- 2) **Corner Analysis:** A ± 10 nm variation in width and thickness produced a group-index spread of only $\pm 0.62\%$, with thickness identified as the dominant sensitivity, demonstrating a robust design.
- 3) **Experimental Verification:** The measured group index ($n_g = 4.32$) agreed with simulation (4.203) to within 2.7%, with the offset consistent with the narrow/thin process corners.
- 4) **Repeatability:** Two nominally identical devices matched to within 0.7% in FSR, confirming fabrication uniformity.

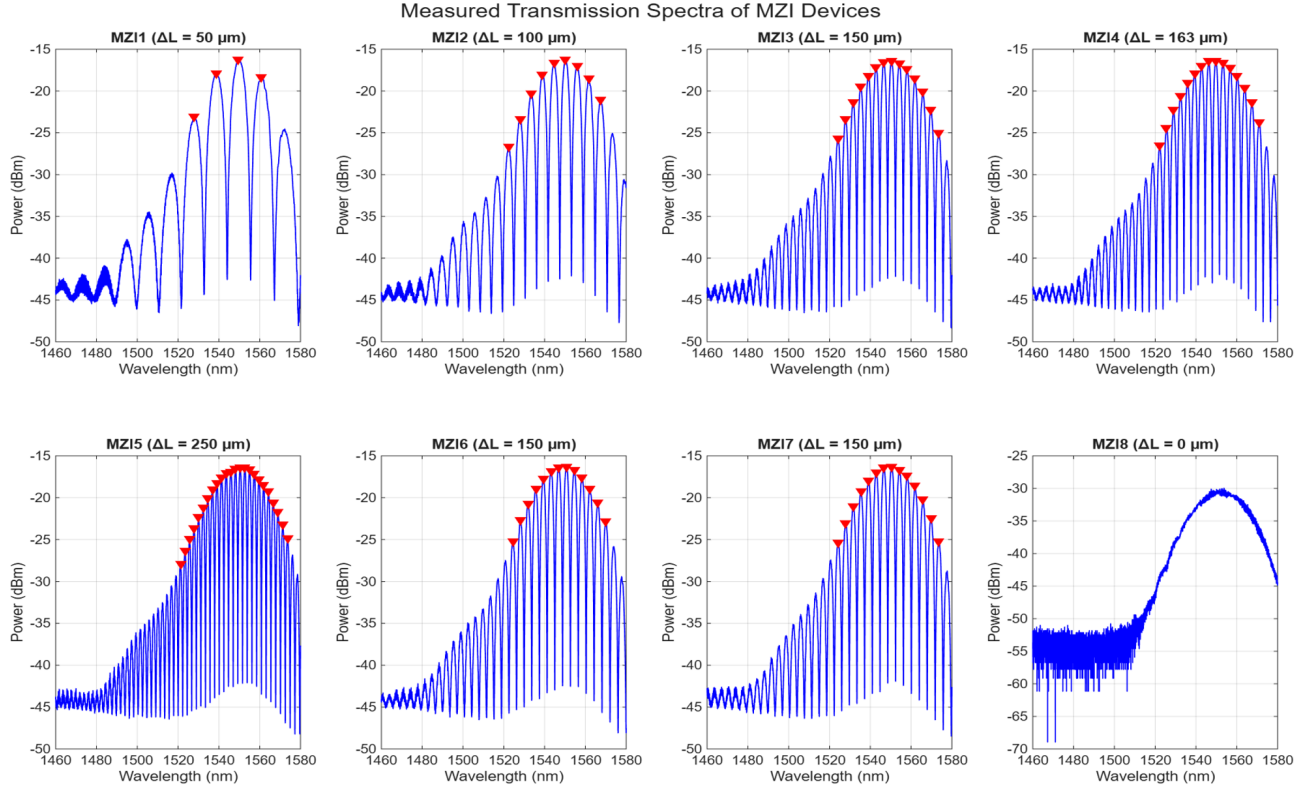


Fig. 9. Measured transmission spectra of the eight fabricated MZI devices. Red markers indicate the automatically detected transmission peaks used for FSR extraction.

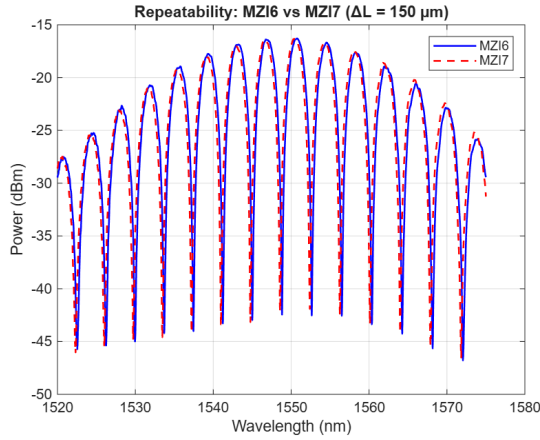


Fig. 11. Overlaid transmission spectra of the two nominally identical devices MZI6 and MZI7 ($\Delta L = 150 \mu\text{m}$), demonstrating fabrication repeatability.

The residual discrepancy between the measured and simulated group index is attributed primarily to systematic deviations between the fabricated and designed waveguide dimensions, and secondarily to peak-detection uncertainty introduced by the grating-coupler envelope, which is most pronounced for the small- ΔL device (MZI1).

IX. CONCLUSION

A complete silicon photonic MZI design workflow was demonstrated, from waveguide eigenmode simulation and compact model extraction through circuit simulation, corner analysis, layout, fabrication, and automated measurement. The linearized FSR analysis provided a consistent group-index cross-check in both simulation and experiment. The fabricated devices showed close agreement with simulation (group index within 2.7%) and excellent repeatability (FSR within 0.7%), while the corner analysis quantified the design's robustness to ± 10 nm manufacturing variation. These results validate the design methodology and establish a foundation for future active devices, such as MZI-based modulators, on the same platform.

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